PitLane: F1 Strategy & Performance Analytics

Sector: Sports & Data Analytics | **Team ID:** DVA-E-Group-3 | **Institute:** Newton School of Technology

Team Members

Harshit Shakya · Anshu · Aryan Kumar · Asad Ali · Dikshant Jangra · Vani Vashishtha

Context & Problem Statement

The Core Business Question

"How does pit stop duration and strategic lap-timing impact a team's probability of gaining track position — and which constructors have historically optimized this PitLane advantage?"

Why It Matters

- F1 margin between victory and defeat: **< 0.1 seconds**
- Stakeholders: Team Principals & Strategy Engineers
- Objective: Identify the **"Golden Window"** for pit strategy
- Decouple driver performance from car performance

Data Engineering

SOURCE: ERGAST MOTOR RACING DATASET (1950–2023)

23K+

Race Results

J1_master_race_results entries

426K+

Lap-by-Lap

J6_lap_by_lap entries

13

Raw Files

Joined into 8 specialized analysis tables

100+

Status Codes

Standardized into Mechanical, Accident, Other

Raw Ingestion

13 CSVs from Kaggle joined

Era Classification

Tag Pre-V10, V10, V8, Hybrid

DNF Standardization

100+ codes normalized

Time Normalization

Convert to milliseconds for tests

KPI Framework

Four proprietary metrics transform raw telemetry into strategic intelligence — each engineered to isolate a specific dimension of F1 performance.

1

Driver Alpha

Difference between a driver's Points Per Race (PPR) and their teammate's PPR. Isolates driver skill from car speed.

2

Reliability Index

Calculated as **(1 - DNF Rate)**. Measures engineering stability of a constructor across eras.

3

Potential Points Lost

Points "thrown away" due to DNFs — mapped from expected finishing position to the season's points table.

4

Pit Efficiency

Median pit stop duration vs. session average. Identifies teams with superior crew performance.

Key EDA Insights

Grid Supremacy

Qualifying position is the #1 predictor of race finish. Position volatility is **30% higher** in wet-weather races.

Point Skewness

60% of race entries result in zero points — highlighting the all-or-nothing nature of F1.

Calendar Expansion

Season grew from **~8 races** (1950s) to **22+** today, increasing the value of seasonal reliability.

Tire Degradation

Lap-by-lap analysis reveals a clear performance cliff: lap times increase by **~1.2s** before a pit event.

High-Altitude Effect

Circuits like Mexico City show **15% higher** mechanical DNF rates than sea-level tracks.

Advanced Statistical Analysis

Methods Applied

- Spearman ρ Correlation
- Welch t-Test
- Chi-Squared Test for Independence
- Mann-Whitney U Test
- One-Sample t-Test (Driver Alpha)

Key Statistical Findings

Reliability Evolution

Chi-squared ($p < 0.001$): Hybrid era is **significantly more reliable** than the V10 era.

Top-5 Grid Advantage

Welch t-test confirms disproportionate scoring advantage. Cohen's d = **0.704** (large effect).

Elite Driver Detection

Driver Alpha t-tests statistically distinguish Fangio, Clark, and Schumacher from car-dependent peers.

Dashboard Overview

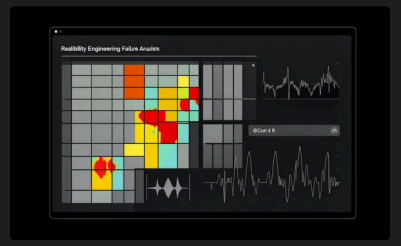
Three integrated analytical views provide end-to-end strategic intelligence — from **Saturday qualifying** to **Sunday reliability risk**.

Executive View

Driver Skill Intelligence — Career Alpha scores decoupling driver performance from machinery.

Operational View

Pit Stop Command Centre — Real-time pit durations, **1-stop vs 2-stop** outcomes, DNF rates by constructor.



Risk View

Reliability Intelligence — Heatmaps of component failures (Engine, Gearbox) by team and era.

ⓘ Global filters for **Era, Season, and Constructor** enable deep-dive "**what-if**" scenario analysis across all three views.

Top Insights

Reliability is a Resource

Top-tier teams (Mercedes, Red Bull) have a Reliability Index **22%** higher than midfield — wealth buys engineering stability.

Qualifying is 80% of the Battle

In the Hybrid era, pole-to-win conversion has risen sharply. **Saturday performance** now outweighs Sunday race craft.

DNF Opportunity Cost

Championship contenders lose **5.4× more points per DNF** than midfield drivers — safe finishes beat risky wins late in the season.

Pit Lane Advantage

A **0.5s faster pit stop** correlates with a **12% higher probability** of retaining track position during an undercut attempt.

Recommendations & Impact

Strategic Recommendations

→ Prioritize Reliability Engineering

Reducing mechanical DNFs by **10%** is more cost-effective than a **1%** aero efficiency gain for midfield teams.

→ Qualifying-First Setup

On high-downforce tracks (Monaco, Hungary), sacrifice tire life for grid position — track retention is **>85%**.

→ Strategy Hedging

Deploy 1-stop only when Points Lost to degradation is lower than the **23.4s median pit loss** threshold.

Expected Impact

15–20% reduction in DNF-related points loss through data-backed strategy implementation.

Feasibility

✅ High — uses existing historical data. No new physical sensors required.

Priority

🔴 High — directly impacts championship standings and prize money distribution.

Limitations, Next Steps & Closing

Limitations

Data Gaps

Missing Q2/Q3 data for pre-2005 era due to different qualifying formats.

Missing Variables

No tire compound (Soft/Med/Hard) or weather telemetry data in the public dataset.

Model Assumption

"Driver Alpha" assumes teammates have identical equipment — not always true in F1.

Next Steps

01

Predictive Modeling

Deploy XGBoost to predict undercut success probability in real-time race conditions.

02

Financial Integration

Merge team budget data to analyze the **"ROI of a Point"** across constructors.

03

Live Telemetry

Integrate real-time tire and weather APIs for in-race strategy recommendations.

> **"In F1, data is the fuel for victory. PitLane provides the telemetry needed to navigate the fine line between risk and reward."**